

B SANJAI

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PROFILE

Results-driven AI & Machine Learning Engineer with expertise in Deep Learning, Generative AI (LLMs), and end-to-end Production Deployment. Proficient in architecting scalable AI applications, building Data Pipelines, and applying Model Optimization for real-time inference. Passionate about leveraging NLP and data-driven approaches to solve complex industrial automation and Cybersecurity challenges.

SKILLS

Languages: Python, C, SQL, JavaScript

AI, Deep Learning & ML Frameworks: PyTorch, Scikit-learn, LangChain, LangGraph, OpenAI, Hugging Face

Backend, Cloud & DevOps: FastAPI, Celery, Docker, Git, Redis, PostgreSQL, MongoDB, ChromaDB, Cloud Services, CI/CD

Frontend & UI: React, Vite, WebSockets, HTML/CSS

Soft Skills: Problem-Solving, Decision-Making, Teamwork, Leadership, Adaptability, Time Management

PROFESSIONAL EXPERIENCE

AI & Machine Learning Intern

08/2025 – Present

IIITDM Kancheepuram

Chennai, India

- Developed an audio-based ML classification pipeline optimized for TinyML inference on STM microcontrollers.
- Built an environment on an electric taxi fleet dataset to evaluate RL dispatch policies (PPO) and forecast battery degradation via time-series models (LSTM, Chronos, TimeFM, TTM).

Networking Intern

05/2025 – 07/2025

Schneider Electric

Chennai, India

- Developed a Python-based network monitoring dashboard to track multiple computers, monitor restart compliance (every two days), and display real-time company system statistics.
- Experienced Cybersecurity threat modeling and vulnerability assessments, securing critical factory telemetry streams and reducing attack surface by 30%.

IIoT & Software Intern

11/2025 – 12/2025

Bray International

Chennai, India

- Automated Data Pipelines using Python and SQL Server via FOCAS2 for CNC machine telemetry, improving data ingestion reliability by 40%.
- Implemented interactive monitoring dashboards to visualize machine states, optimizing production scheduling and reducing unplanned downtime by 15%.

PROJECTS

Agentic Blog Generation System – Autonomous Multi-Language Content Pipeline

- Architected an asynchronous multi-agent Natural Language Processing pipeline using **LangGraph**, **FastAPI**, and **Celery**, automating end-to-end content generation and reducing production time by 90%.
- Engineered real-time state streaming via **Redis** Pub/Sub and **WebSockets**, integrated **ChromaDB** for RAG context retrieval, and deployed to Cloud Services via CI/CD.

Real-Time API Traffic Anomaly Detection & AIOps Dashboard

- Developed a full-stack observability dashboard deployed on **Hugging Face Spaces** via **Docker**, utilizing a single **FastAPI** process to serve both a REST/WebSocket API and a compiled **React** UI, backed by **Neon DB** for persistent storage.
- Engineered a real-time multivariate anomaly ensemble (**Isolation Forest**, IQR, Z-score) processing live API metrics, integrated with **Groq** and **OpenRouter LLMs** for automated root-cause analysis and mitigation plans.

EDUCATION

B.Tech Electronics and Communication Engineering

08/2023 – Present

Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Kancheepuram

Chennai, India

CGPA: 8.27

CERTIFICATES

Workshop on Machine Learning, Data Science: Techniques and Trends

NIT Trichy

Complete Generative AI Course With Langchain and Huggingface

Krish Naik

End-to-End Data Science, Machine Learning, Deep Learning, NLP Projects

Krish Naik